

1. both arrays and stacks are linear data structures used to store collections of elements. Both can be implemented using contiguous memory locations
2. A stack is useful in compiler software for parsing expressions, managing function calls, and handling local variables.
3. 8 13
12 12
4. The hot plate problem is analogous to a stack because it follows the LIFO principle
5. 5 8
13 12
6. FIFO process elements in the order they were added, while LIFO processes elements in the reverse order they were added
7. A LIFO structure, commonly known as a stack, processes the most recently added element first.

A FIFO structure, known as a queue, processes the oldest (first) element first.

8. A. false a stack data structure does not have a front and rear only a top
B. True
C. False. In a stack top refers to the last item pushed onto a stack
D. False. The isEmpty operation returns a boolean value not an integer
E. true
F. False in a queue all removals are made at the front

G. False. The enqueue operation adds an item to the rear of the queue

H.true

I. True

J. True

K. True